

Subset Notation

Set Theory Worksheet · Grade 6–8

Name: _____

Date: _____

Learning Objectives

- Use subset and not-a-subset symbols to describe relationships between sets
- List all distinct subsets of a given set, including the empty set
- Use the formula 2^n to count the number of distinct subsets of a set with n elements

Problems

1. Set $A = \{2, 4, 6\}$ and Set $B = \{2, 4, 6, 8, 10\}$. Is Set A a subset of Set B ? Write the correct notation.

$A \subseteq B$ or $A \not\subseteq B$

2. Set $A = \{m, a, t, h\}$ and Set $B = \{m, a, t\}$. Is Set A a subset of Set B ? Write the correct notation.

$A \subseteq B$ or $A \not\subseteq B$

3. Write TRUE or FALSE: The empty set (null set) is always a subset of any set.

$\emptyset \subseteq B$ for any set B

4. Set $A = \{x \text{ such that } x \text{ is a letter in the word 'CAT'}\}$ and Set $B = \{x \text{ such that } x \text{ is a letter in the word 'CATCH'}\}$. List the elements of each set, then determine if A is a subset of B .

$\left\{ \begin{array}{l} A = \{C, A, T\} \\ B = \{C, A, T, H\} \end{array} \right.$

5. List ALL distinct subsets of the set $\{5\}$.

$\{5\}$



6. List ALL distinct subsets of the set {1, 2, 3}.

{1, 2, 3}

7. How many distinct subsets does a set with 4 elements have? Use the formula and show your work.

2^n

8. Set A = {a, e, i, o, u} and Set B = {a, b, c, d, e}. Is A a subset of B? Is B a subset of A? Write the correct notation for each.

$\left\{ \begin{array}{l} A \subseteq B \text{ or } A \not\subseteq B \\ B \subseteq A \text{ or } B \not\subseteq A \end{array} \right.$

9. A set has 5 elements. How many of its distinct subsets contain exactly 2 elements? (Hint: list or use combinations.)

$\binom{5}{2}$

10. Set A = {x such that x is a whole number and $1 \leq x \leq 4$ } and Set B = {x such that x is an even whole number and $1 \leq x \leq 10$ }. Write both sets in roster form, determine if $A \subseteq B$, and find the total number of distinct subsets of Set A.

$\left\{ \begin{array}{l} A = \{1, 2, 3, 4\} \\ B = \{2, 4, 6, 8, 10\} \end{array} \right.$



Subset Notation — Answer Key

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Answer Key

1. Answer: $A \subseteq B$ (A is a subset of B)

- Check if every element of A is also in B.
- $2 \in B$ ✓, $4 \in B$ ✓, $6 \in B$ ✓
- All elements of A are found in B, so $A \subseteq B$.

2. Answer: $A \not\subseteq B$ (A is not a subset of B)

- Check if every element of A is also in B.
- $m \in B$ ✓, $a \in B$ ✓, $t \in B$ ✓, but $h \notin B$ ✗
- Since h is in A but not in B, $A \not\subseteq B$.

3. Answer: TRUE

- By definition, the empty set has no elements.
- Since there are no elements to violate the subset condition, the empty set is a subset of every set.
- The statement is TRUE.

4. Answer: $A \subseteq B$

- List the distinct letters: $A = \{C, A, T\}$ and $B = \{C, A, T, H\}$.
- Check: $C \in B$ ✓, $A \in B$ ✓, $T \in B$ ✓
- All elements of A are in B, so $A \subseteq B$.

5. Answer: $\{5\}$ and $\emptyset$ — 2 subsets total

- A set with 1 element has $2^1 = 2$ distinct subsets.
- Subset 1: $\{5\}$ (the set itself)
- Subset 2: $\emptyset$ (the empty set)
- Total: 2 distinct subsets.

6. Answer: $\{1,2,3\}$, $\{1,2\}$, $\{1,3\}$, $\{2,3\}$, $\{1\}$, $\{2\}$, $\{3\}$, $\emptyset$ — 8 subsets

- A set with 3 elements has $2^3 = 8$ distinct subsets.
- 3-element subset: $\{1, 2, 3\}$
- 2-element subsets: $\{1,2\}$, $\{1,3\}$, $\{2,3\}$
- 1-element subsets: $\{1\}$, $\{2\}$, $\{3\}$
- 0-element subset: $\emptyset$
- Total: 8 distinct subsets.

7. Answer: 16 subsets

- The number of distinct subsets of a set with n elements is 2^n .
- Here $n = 4$, so the number of subsets = 2^4 .
- $2^4 = 16$.
- There are 16 distinct subsets



8. Answer: $A \not\subseteq B$ and $B \not\subseteq A$

- Check $A \subseteq B$: $1 \notin B$, so $A \not\subseteq B$.
 - Check $B \subseteq A$: $6 \notin A$, so $B \not\subseteq A$.
 - This confirms subset notation is NOT commutative.
 - Both $A \not\subseteq B$ and $B \not\subseteq A$.
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9. Answer: 10 subsets

- We need to choose 2 elements from 5, where order does not matter.
 - Use the combination formula: $C(5,2) = 5! / (2! \times 3!)$
 - $C(5,2) = (5 \times 4) / (2 \times 1) = 20 / 2 = 10$.
 - There are 10 distinct 2-element subsets.
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10. Answer: $A \not\subseteq B$; Set A has 16 distinct subsets

- List A in roster form: $A = \{1, 2, 3, 4\}$.
 - List B in roster form: $B = \{2, 4, 6, 8, 10\}$.
 - Check $A \subseteq B$: $1 \notin B$ and $3 \notin B$, so $A \not\subseteq B$.
 - Count distinct subsets of A: $n = 4$, so $2^4 = 16$ distinct subsets.
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